

NANDI INSTITUTE OF MANGEMENT & SCIENCE



MINI PROJECT

On

“BUS TRAIN RESERVATION SYSTEM”

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COURSE&SEM: BCA &3rd SEM

DEPARTMENT OF BCA 2025-26

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CERTIFICATE

This is to certify that FATHIMA KHUSHBU bearing UN16NB24S0148 has satisfactorily completed the mini project “BUS TRAIN RESERVATION SYSTEM” in the academic year 2025-26. This project work has been carried out per the curriculum requirement of NIMS for BCA 3rd SEMESTER, under guidance and supervision.

INTERNAL GUIDE :Prof. PRIYANKA. S

COURSE &SEM: BCA&3rd SEM

SIGNATURE

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ACKNOWLEDGEMENT

“Task successful” makes everyone happy. But the happiness will be gold without ginner if we didn’t state the person who have supported us to make it is success.

Success will be crowned to people who makes it a really but the people who constant guidance and encouragement made it possible will be crowed first on the eve success.

This acknowledgment transcends the reality of formality when we would like to express deep gratitude and respect to all those people behind the screen who guided, inspired and helped me for the completion of my mini project

I consider myself lucky enough to get such a good project. This project would add as an assent to my academic profile.

PRIYANKA.S Assistant professor, Department of BCA of her constant support and co- operation during the mini project.

Fathima Khushbu

U16NB24S0148

OBJECTIVE

The objective of web design is to create visually appealing, user-friendly, and responsive websites that effectively communicate a brand's identity, enhance user experience, and achieve business goals. It involves combining creativity and technical skills to develop engaging, accessible, and SEO-friendly websites that support online marketing strategies, improve customer engagement, and strengthen a company's digital presence. The ultimate aim is to deliver professional web design solutions that balance aesthetics and functionality to drive user interaction and measurable business results.

SOFTWARE REQUIREMENTS

COMPONENT	REQUIRMENTS
Operating System	Windows
Editor/IDE	Visual studio code
Browser	Google chrome
Languages	HTML, CSS, JS

PROBLEM STATEMENT

Many businesses and individuals struggle to establish a strong online presence due to poorly designed websites that are difficult to navigate, visually unappealing, or not optimized for different devices. These issues lead to low user engagement, reduced customer trust, and missed business opportunities. Therefore, there is a need for an effective web design solution that focuses on creating responsive, accessible, and aesthetically pleasing websites that provide a seamless user experience, clearly communicate information, and help achieve organizational or personal goals.

ALGORITHM

Step 1: Start.

Step 2: Identify the website's purpose — to allow users to search, view, and book bus/train tickets online.

Step 3: Gather front-end requirements (e.g., home page, search form, schedule display, booking form, payment interface, and confirmation page).

Step 4: Design wireframes or mockups for each page layout.

Step 5: Create the website layout using **HTML** for structure, **CSS** for styling, and **JavaScript** for interactivity and form validation.

Step 6: Implement the following main pages

- **Home Page:** Provides navigation and introduction.
- **Search Page:** Allows users to enter source, destination, and travel date.
- **Result Page:** Displays available buses/trains with details and a “Book Now” option.
- **Booking Page:** Collects passenger details and payment info.
- **Confirmation Page:** Shows ticket confirmation and booking summary.

Step 7: Make the design **responsive** for different devices.

Step 8: Test all front-end features (navigation, forms, validation, responsiveness).

Step 9: Optimize loading speed and fix UI issues.

Step 10: Deploy the front-end design.

Step 11: End.

SOURCE CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Cinematic 3D Bus & Train Reservation</title>
<style>
* { box-sizing: border-box; }
body {
    margin: 0;
    font-family: "Poppins", sans-serif;
    color: #fff;
    overflow-x: hidden;
    background: #000;
    perspective: 1000px;
}
/* Fullscreen slideshow background */
#bg-slideshow {
    position: fixed;
    top: 0; left: 0;
    width: 100%;
    height: 100%;
    background-size: cover;
    background-position: center;
    animation: slideshow 20s infinite;
    z-index: -3;
    filter: brightness(0.5);
}
@keyframes slideshow {
    0% { background-image: url('https://images.unsplash.com/photo-1525609004556-c46c7d6cf023?auto=format&fit=crop&w=1920&q=80'); }
    25% { background-image: url('https://images.unsplash.com/photo-1499856871958-5b9627545d1a?auto=format&fit=crop&w=1920&q=80'); }
```

```
50% { background-image: url('https://images.unsplash.com/photo-1517154421773-0529f29ea451?auto=format&fit=crop&w=1920&q=80'); }
```

```
75% { background-image: url('https://images.unsplash.com/photo-1504384308090-c894fdcc538d?auto=format&fit=crop&w=1920&q=80'); }
```

```
100% { background-image: url('https://images.unsplash.com/photo-1525609004556-c46c7d6cf023?auto=format&fit=crop&w=1920&q=80'); }
```

```
}
```

```
/* Gradient overlay */
```

```
#overlay {
```

```
  position: fixed;
```

```
  top: 0; left: 0;
```

```
  width: 100%; height: 100%;
```

```
  background: linear-gradient(135deg, rgba(0,0,0,0.6), rgba(0, 30, 80, 0.7), rgba(60, 0, 90, 0.6));
```

```
  z-index: -2;
```

```
}
```

```
header {
```

```
  text-align: center;
```

```
  padding: 1.2rem;
```

```
  font-size: 2rem;
```

```
  letter-spacing: 1px;
```

```
  background: rgba(0,0,0,0.5);
```

```
  font-weight: 600;
```

```
  color: #00e5ff;
```

```
  text-shadow: 0 0 20px #00e5ff;
```

```
}
```

```
.container {
```

```
  max-width: 500px;
```

```
  margin: 40px auto;
```

```
  background: rgba(255,255,255,0.12);
```

```
  padding: 2rem;
```

```
  border-radius: 20px;
```

```
  backdrop-filter: blur(15px);
```

```
  box-shadow: 0 0 25px rgba(0,255,255,0.4);
```

```
border-top: 6px solid #00e5ff;
border-radius: 50%;
width: 60px; height: 60px;
animation: spin 1s linear infinite;
margin: 20px auto;
}
@keyframes spin { to { transform: rotate(360deg); } }
.ticket {
background: rgba(255,255,255,0.2);
padding: 1rem;
border-radius: 10px;
margin-top: 1rem;
font-size: 0.95rem;
}
.logout-btn {
background: linear-gradient(90deg, #ff5252, #ff1744);
color: white;
margin-top: 10px;
}
@media(max-width: 500px) {
.seat-grid { grid-template-columns: repeat(3, 55px); gap: 10px; }
}
</style>
</head>
<body>
<div id="bg-slideshow"></div>
<div id="overlay"></div>
<header>3D Bus & Train Reservation Portal</header>
<!-- LOGIN -->
<div class="container" id="loginPage">
<h2>Login</h2>
<input type="text" id="username" placeholder="Username" />
```



```
<input type="password" id="password" placeholder="Password" />

<button onclick="login()">Login</button>

function renderSeats() {
  seatGrid.innerHTML = "";
  for (let i = 1; i <= 20; i++) {
    const seat = document.createElement("div");
    seat.textContent = i;
    seat.classList.add("seat");
    if (bookedSeats.includes(i)) seat.classList.add("booked");
    seat.style.animationDelay = `${i * 0.05}s`;
    seat.onclick = () => toggleSeat(i, seat);
    seatGrid.appendChild(seat);
  }
}

function toggleSeat(num, el) {
  if (el.classList.contains("booked")) return alert("Seat already booked!");
  if (selectedSeats.includes(num)) {
    selectedSeats = selectedSeats.filter(s => s !== num);
    el.classList.remove("selected");
  } else {
    selectedSeats.push(num);
    el.classList.add("selected");
  }
  fareInfo.textContent = `Selected: ${selectedSeats.length} | Total: ₹${selectedSeats.length * 250}`;
}

function proceedToPay() {
  if (!selectedSeats.length) return alert("Select at least one seat!");
  reservationPage.classList.add("hidden");
  paymentPage.classList.remove("hidden");
  setTimeout(() => {
    paymentPage.classList.add("hidden");
  }, 1000);
}
```

```
successPage.classList.remove("hidden");  
confirmBooking();  
}, 2500);  
}
```

```
function confirmBooking() {  
  const t = {  
    mode: mode.value,  
    name: transport.value,  
    from: from.value,  
    to: to.value,  
    date: date.value,  
    seats: [...selectedSeats],  
    fare: selectedSeats.length * 250,  
    paymentId: "TXN" + Math.floor(Math.random() * 1000000)  
  };  
  tickets.push(t);  
  bookedSeats.push(...selectedSeats);  
  ticketDetails.innerHTML = `  
    <div class="ticket">  
      <strong>${t.mode}</strong> ${t.name}<br>  
      ${t.from} → ${t.to}<br>  
      Date: ${t.date}<br>  
      Seats: ${t.seats.join(", ")}<br>  
      Total Fare: ₹${t.fare}<br>  
      Payment ID: ${t.paymentId  
    </div>`;  
  selectedSeats = [];  
  renderSeats();  
}  
  
function showPrevious() {  
  reservationPage.classList.add("hidden");
```

```

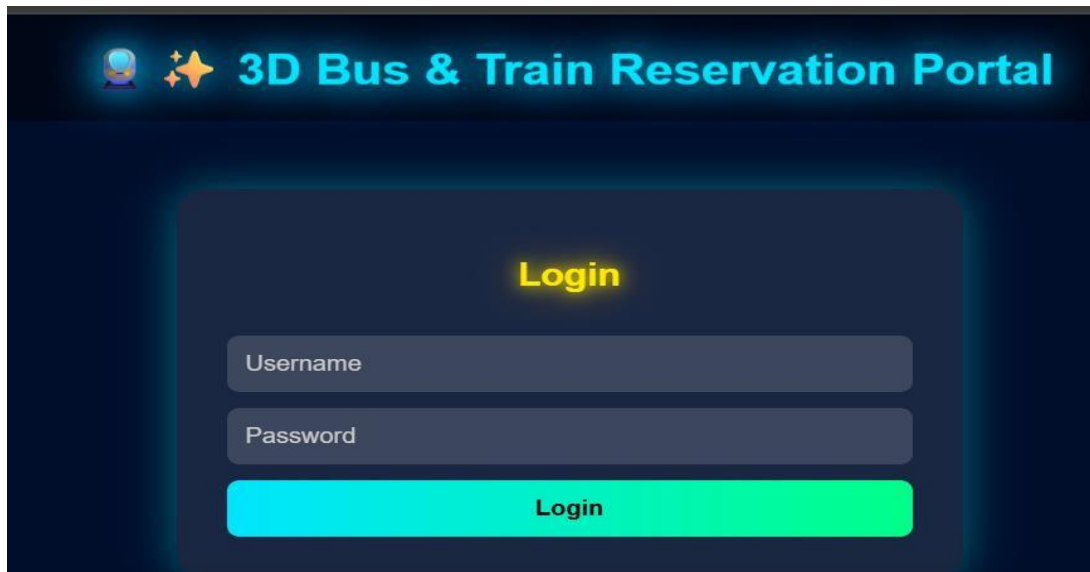
    successPage.classList.add("hidden");
    previousPage.classList.remove("hidden");
    previousTickets.innerHTML = tickets.length ? "" : "<p>No previous tickets yet.</p>";
    tickets.forEach(t => {
        previousTickets.innerHTML += `
            <div class="ticket">
                <strong>${t.mode}</strong> ${t.name}<br>
                ${t.from} → ${t.to}<br>
                Date: ${t.date}<br>
                Seats: ${t.seats.join(", ")}<br>
                Fare: ₹${t.fare}<br>
                Payment ID: ${t.paymentId}
            </div>`;
    });
}

function backToBooking() {
    previousPage.classList.add("hidden");
    successPage.classList.add("hidden");
    reservationPage.classList.remove("hidden");
    renderSeats();
    fareInfo.textContent = `Selected: 0 | Total: ₹0`;
}

function logout() {
    document.querySelectorAll('.container').forEach(c => c.classList.add('hidden'));
    loginPage.classList.remove('hidden');
    selectedSeats = [];
}
</script>
</body>
</html>

```

OUTPUT



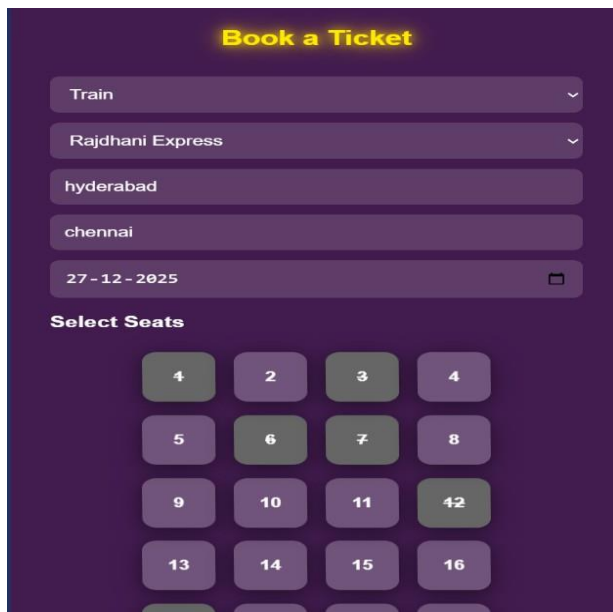
The login screen features a dark blue background with a glowing effect. At the top, there is a header with a user icon, a star icon, and the text "3D Bus & Train Reservation Portal". Below this, the word "Login" is displayed in a glowing yellow font. There are two input fields: "Username" and "Password", both with a glowing blue border. A glowing blue "Login" button is positioned at the bottom.

Login

Username

Password

Login



The "Book a Ticket" screen has a dark purple background. It includes a "Book a Ticket" header in glowing yellow. Below the header, there are four input fields: "Train" (with a dropdown arrow), "Rajdhani Express" (with a dropdown arrow), "hyderabad", and "chennai". A date field shows "27-12-2025" with a calendar icon. Below these fields is a "Select Seats" section with a grid of 16 buttons numbered 1 to 16. Buttons 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16 are visible.

Book a Ticket

Train

Rajdhani Express

hyderabad

chennai

27-12-2025

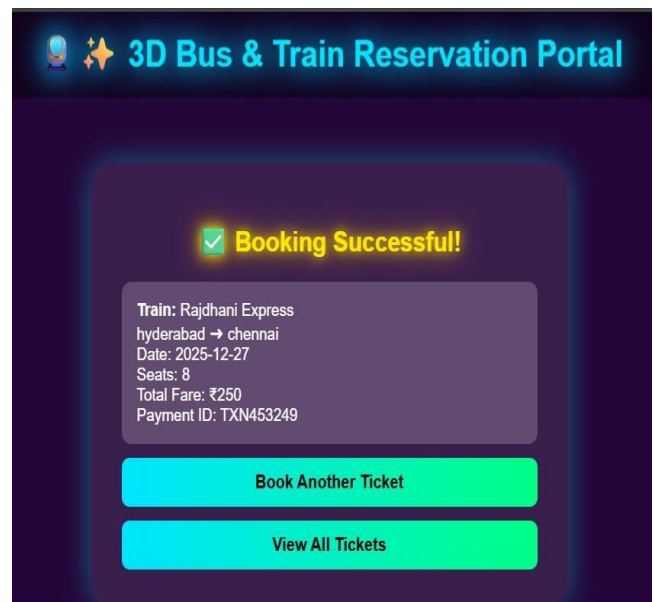
Select Seats

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16



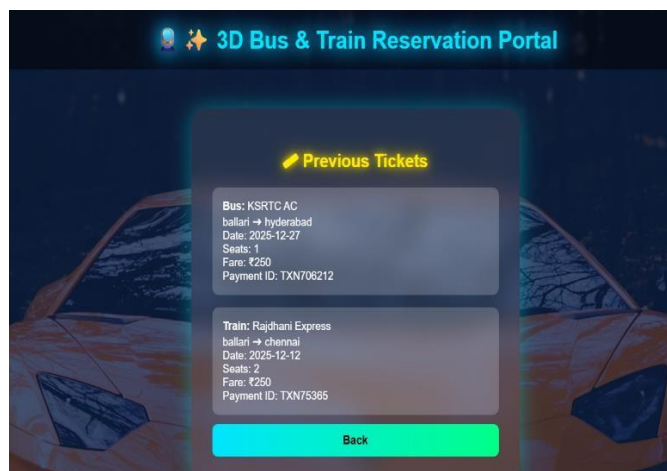
The "Booking Successful" screen has a dark purple background. It features a header with a user icon, a star icon, and the text "3D Bus & Train Reservation Portal". Below this, the text "Booking Successful!" is displayed in a glowing yellow font with a checkmark icon. There is a box containing the booking details: "Train: Rajdhani Express", "hyderabad → chennai", "Date: 2025-12-27", "Seats: 8", "Total Fare: ₹250", and "Payment ID: TXN453249". Below this box are two glowing blue buttons: "Book Another Ticket" and "View All Tickets".

Booking Successful!

Train: Rajdhani Express
hyderabad → chennai
Date: 2025-12-27
Seats: 8
Total Fare: ₹250
Payment ID: TXN453249

Book Another Ticket

View All Tickets



The "Previous Tickets" screen has a dark blue background with a glowing effect. It features a header with a user icon, a star icon, and the text "3D Bus & Train Reservation Portal". Below this, the text "Previous Tickets" is displayed in a glowing yellow font with a key icon. There are two boxes containing ticket details. The first box shows: "Bus: KSRTC AC", "ballari → hyderabad", "Date: 2025-12-27", "Seats: 1", "Fare: ₹250", and "Payment ID: TXN706212". The second box shows: "Train: Rajdhani Express", "ballari → chennai", "Date: 2025-12-12", "Seats: 2", "Fare: ₹250", and "Payment ID: TXN75385". Below these boxes is a glowing blue "Back" button.

Previous Tickets

Bus: KSRTC AC
ballari → hyderabad
Date: 2025-12-27
Seats: 1
Fare: ₹250
Payment ID: TXN706212

Train: Rajdhani Express
ballari → chennai
Date: 2025-12-12
Seats: 2
Fare: ₹250
Payment ID: TXN75385

Back

CONCLUSION

The development of a Bus and Train Reservation System with a web-based front end represents a significant step toward modernizing public transportation services, providing passengers with an efficient, accessible, and user-friendly platform for managing travel bookings. By employing modern web design principles, the system ensures an intuitive interface that allows users to easily navigate through schedules, seat availability, fare details, and reservation options, facilitating a seamless booking experience from any device with internet access. This eliminates the need for physical ticket counters and reduces reliance on manual processes, saving time and effort for both passengers and transport operators while minimizing errors and miscommunication. The integration of the front-end interface with a robust backend ensures real-time updates on seat availability, cancellations, and modifications, which enhances operational reliability and ensures that passengers receive accurate information instantly. Security considerations, such as encrypted transactions, secure user authentication, and protection of personal and payment data, further strengthen user confidence in online booking systems.

REFERENCE

- Duckett, J. *HTML & CSS: Design and Build Websites*. Wiley, 2011.
- Mozilla Developer Network (MDN) – HTML, CSS, and JavaScript documentation:
<https://developer.mozilla.org>